Cloud-Based Honeypot Deployment and SOC Threat Monitoring Using Microsoft Sentinel

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Project Type: Cloud-Based Honeypot Security Monitoring
Platform: Microsoft Azure & Microsoft Sentinel
Monitoring Duration: 2 Weeks
Virtual Machine: CORP-CLIENT-DATABASE-EAST-1
Region: East US
Log Analytics Workspace: LOG-soc-lab-0000
Operating System: Windows
Date: March 30, 2026

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1. EXECUTIVE SUMMARY

This project presents the design, deployment, and operation of a cloud-based honeypot environment using Microsoft Azure and Microsoft Sentinel to simulate real-world Security Operations Center (SOC) activities.

A deliberately exposed Windows virtual machine was configured to attract malicious actors via Remote Desktop Protocol (RDP). Over a two-week monitoring period, the system captured and analyzed real attack traffic, primarily brute force login attempts originating from multiple geographic locations.

Microsoft Sentinel was used as the SIEM platform to ingest logs, detect threats, generate incidents, and support investigation and response workflows. Detected threats were enriched using external threat intelligence sources, and malicious IP addresses were mitigated using Azure Network Security Group (NSG) rules.

Beyond basic monitoring, this project demonstrates deeper SOC capabilities including detection logic design, attacker behavior analysis, threat hunting, alert tuning, and response optimization. The findings highlight the risks of exposed services and the importance of proactive defense strategies.

2. INTRODUCTION

Cloud environments are continuously targeted by attackers scanning for exposed services such as RDP. Misconfigured systems are often identified and attacked within minutes of exposure.

This project simulates a real-world SOC environment by deploying a honeypot designed to attract malicious activity. The goal is to observe attacker behavior, analyze attack patterns, and develop detection and response strategies based on real telemetry.

3. OBJECTIVES

- Deploy a cloud-based honeypot environment
- Monitor real-time attacker activity
- Detect brute force and unauthorized login attempts
- Perform detailed incident investigation and analysis
- Execute incident response actions
- Conduct threat hunting activities
- Visualize attack patterns
- Simulate real SOC analyst workflows

4. METHODOLOGY

The project followed a structured SOC lifecycle approach:

Phase 1: Environment Deployment

- Created Windows VM in Azure
- Exposed RDP (Port 3389) to the internet
- Connected VM to Log Analytics Workspace

Phase 2: Log Collection

- Enabled SecurityEvent logs
- Integrated Microsoft Sentinel

Phase 3: Detection Engineering

- Developed KQL-based detection rules

Phase 4: Monitoring & Analysis

- Monitored logs over a two-week period

Phase 5: Incident Investigation

- Conducted timeline and behavioral analysis

Phase 6: Threat Hunting

- Performed proactive log searches to identify hidden patterns

Phase 7: Threat Intelligence

- Validated indicators using external platforms

Phase 8: Incident Response

- Blocked malicious IPs using NSG rules

Phase 9: Reporting & Visualization

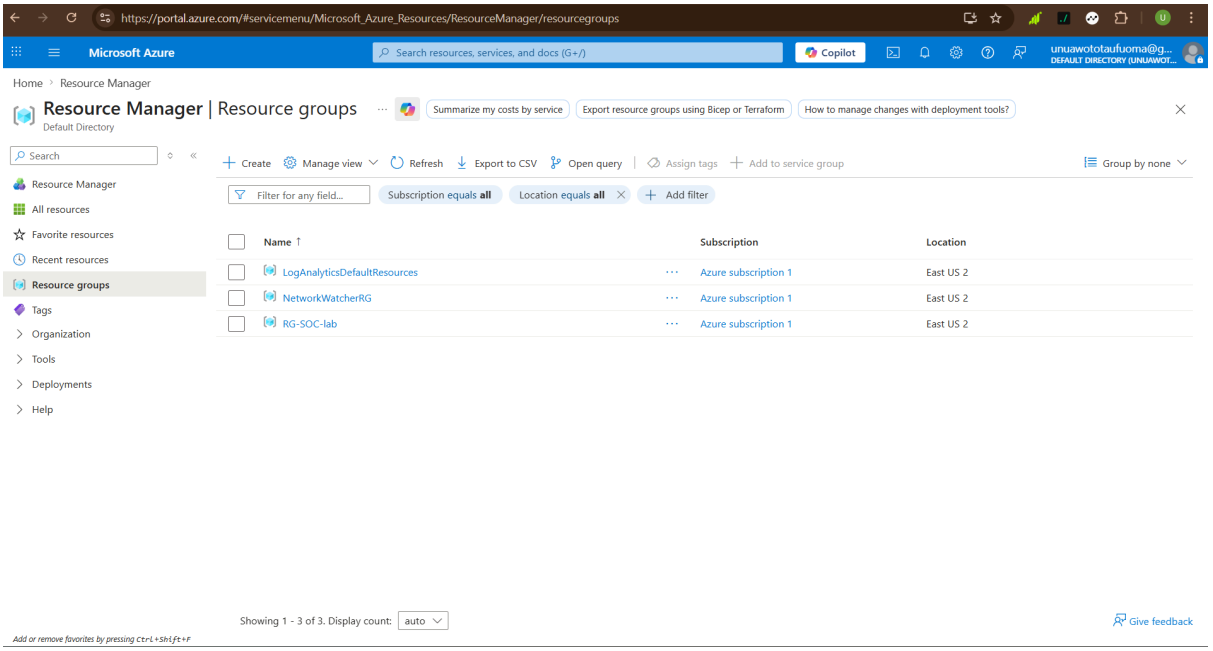
- Built dashboards using Sentinel Workbooks

5. ENVIRONMENT ARCHITECTURE

The environment consisted of:

- Azure Virtual Machine (Windows)
- Microsoft Sentinel (SIEM)
- Log Analytics Workspace
- Azure Network Security Group (NSG)

Figure 1: Azure Architecture Overview



Description: Displays all Azure resources and their relationships within the environment.

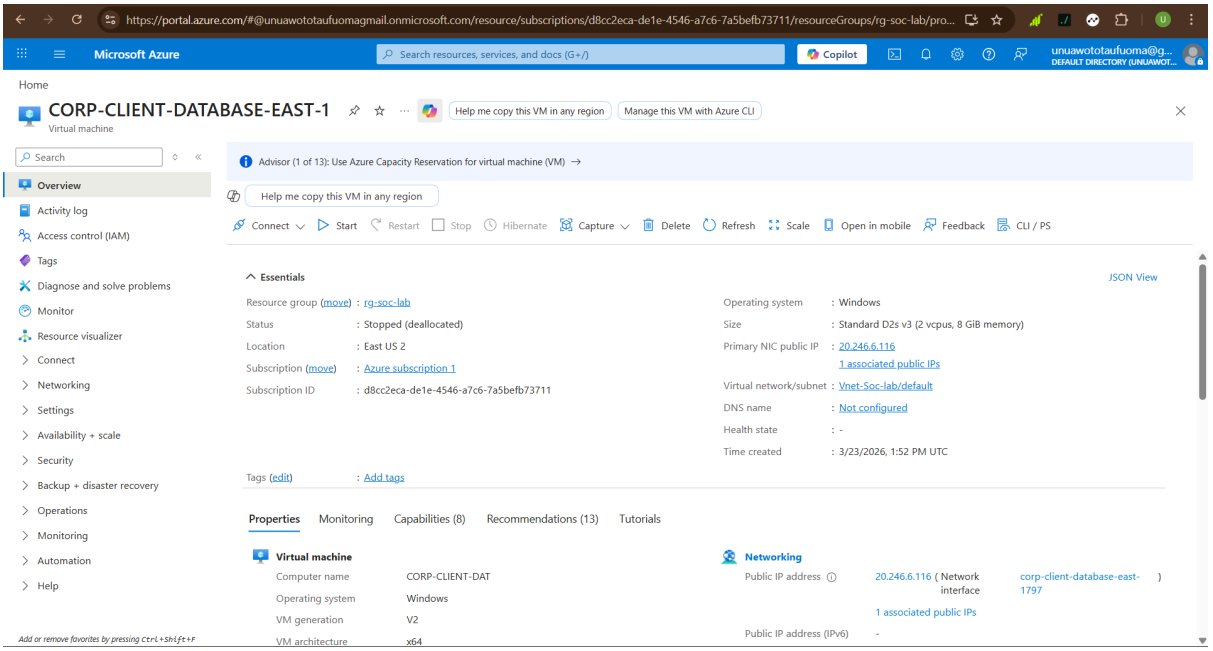
6. TOOLS & TECHNOLOGIES

- Microsoft Azure
- Microsoft Sentinel
- Log Analytics Workspace
- Kusto Query Language (KQL)
- Windows Security Logs
- Azure NSG
- VirusTotal

7. HONEYPOT CONFIGURATION

- VM Name: CORP-CLIENT-DATABASE-EAST-1
- RDP exposed to the internet
- No restrictive inbound filtering
- Continuous monitoring enabled

Figure 2: VM Configuration



Description: Shows configuration of the honeypot VM.

8. LOG COLLECTION & MONITORING

Collected logs:

- Event ID 4625 (Failed Login)
- Event ID 4624 (Successful Login)

Captured data:

- Source IP
- Timestamp
- Username attempts
- Geographic origin

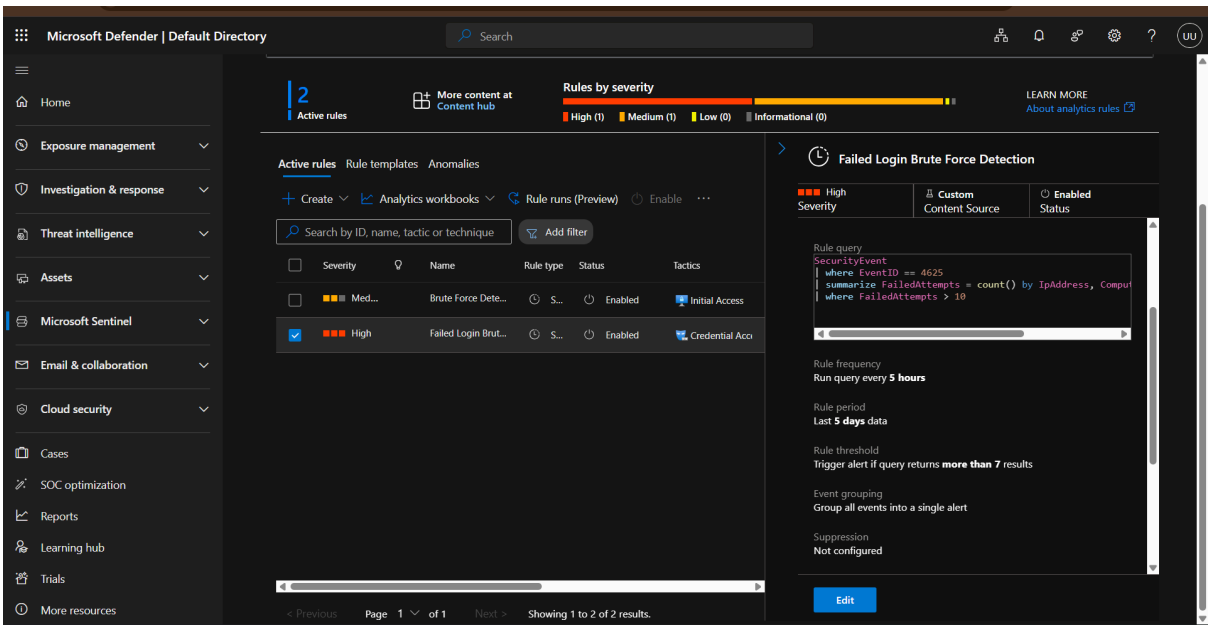
9. DETECTION ENGINEERING

Detection rules were designed to identify suspicious login behavior.

Design Logic:

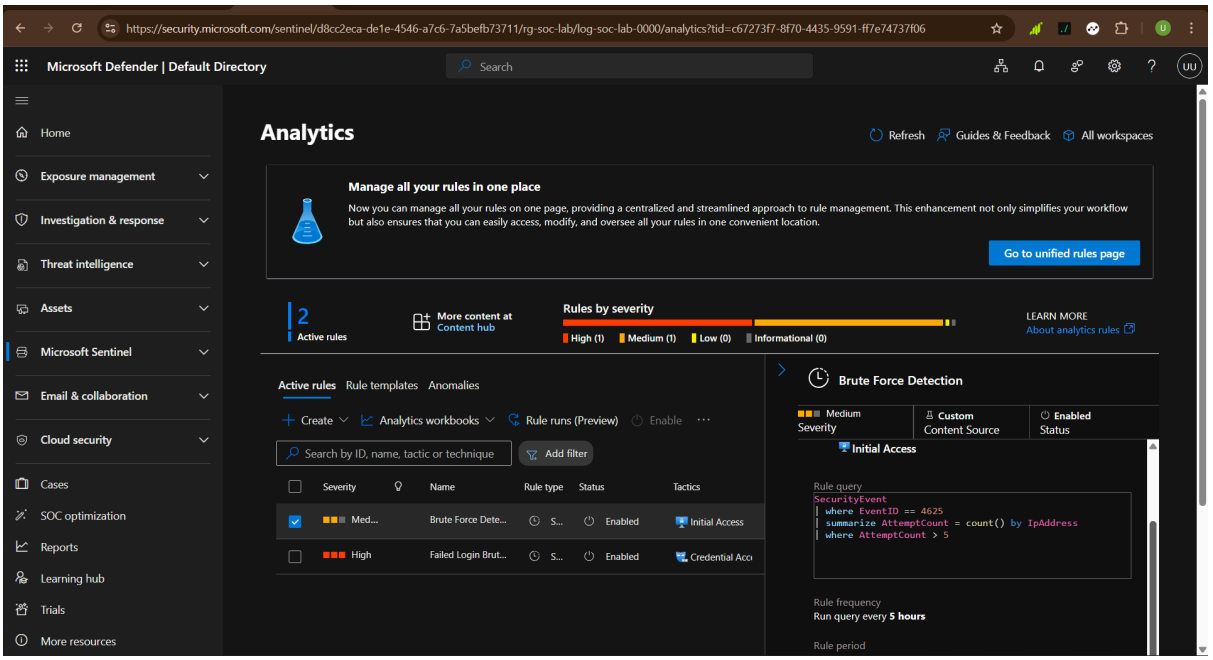
- Repeated failed logins from a single IP indicate brute force attempts
- Thresholds were set to reduce false positives from normal user behavior

Figure 3: Failed Login Detection Rule



Description: KQL query detecting multiple failed login attempts within a defined time window.

Figure 4: Brute Force Detection Rule



Description: Detects high-frequency login attempts from a single IP.

Detection Considerations:

- Threshold tuning to avoid alert fatigue
- Potential false positives from legitimate users

- Need for contextual enrichment

10. MITRE ATT&CK MAPPING

The observed attacks align with the following MITRE ATT&CK techniques:

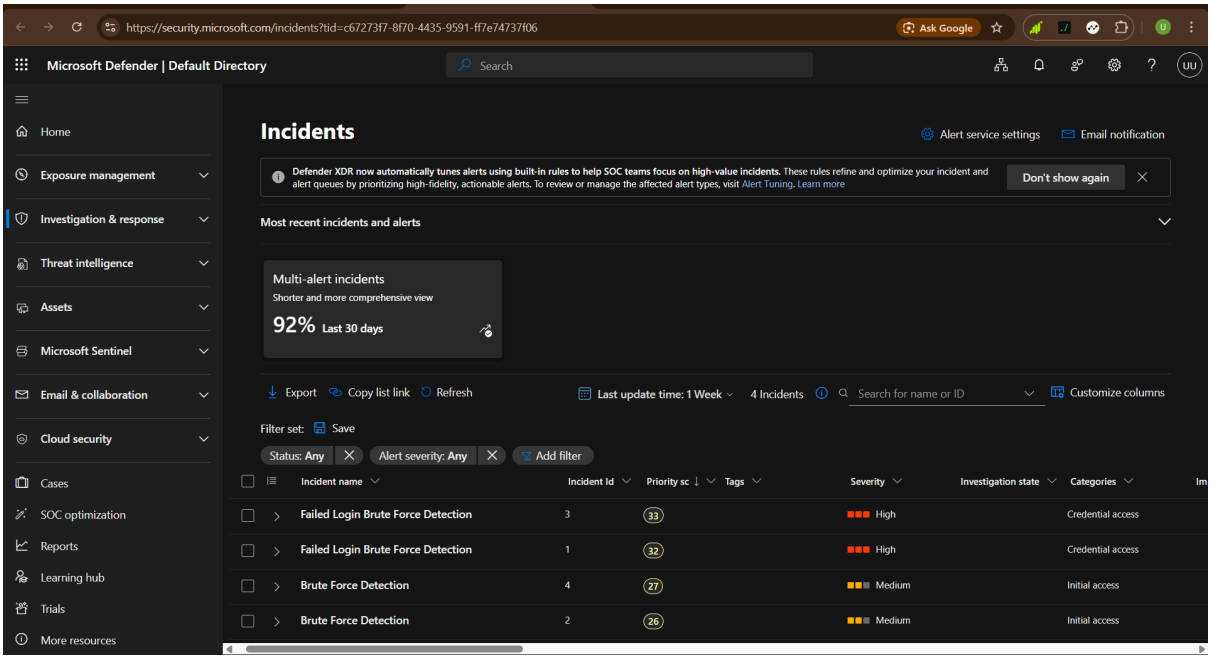
- **T1110 – Brute Force**
- **T1021 – Remote Services (RDP)**

This mapping aligns observed activity with known attacker techniques.

11. MONITORING RESULTS

- Total Incidents: 12
- Brute Force Attacks: 6
- Failed Login Incidents: 6
- Highest Attempts from One IP: 14,738
- Top Attacking Country: China

Figure 5: Sentinel Incidents Dashboard



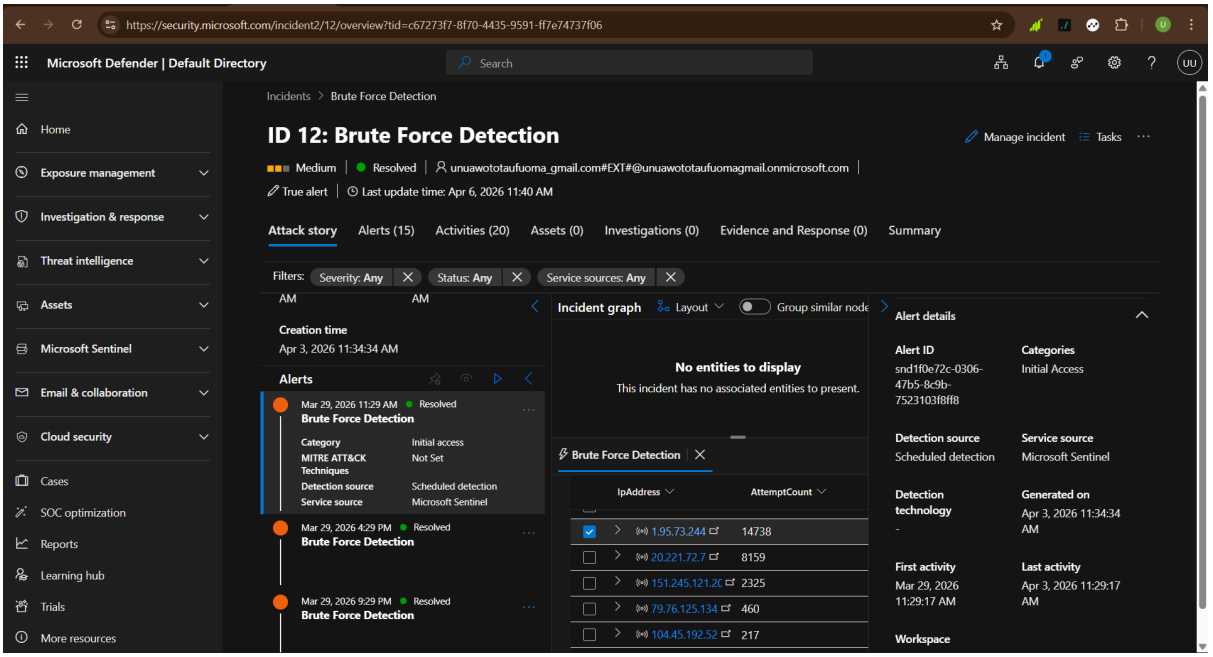
Description: Displays detected incidents and alert distribution.

12. DETAILED INCIDENT ANALYSIS

Incident: RDP Brute Force Attack

- Attacker IP: 1.95.73.244
- Country: China
- Attempts: 14,738
- Severity: High

Figure 6: Incident Overview



Description: Shows timeline and login attempt distribution.

Behavioral Analysis:

- Continuous login attempts over extended periods
- Activity peaks observed during specific time windows
- Targeted usernames included common administrative accounts (e.g., admin, guest, user)

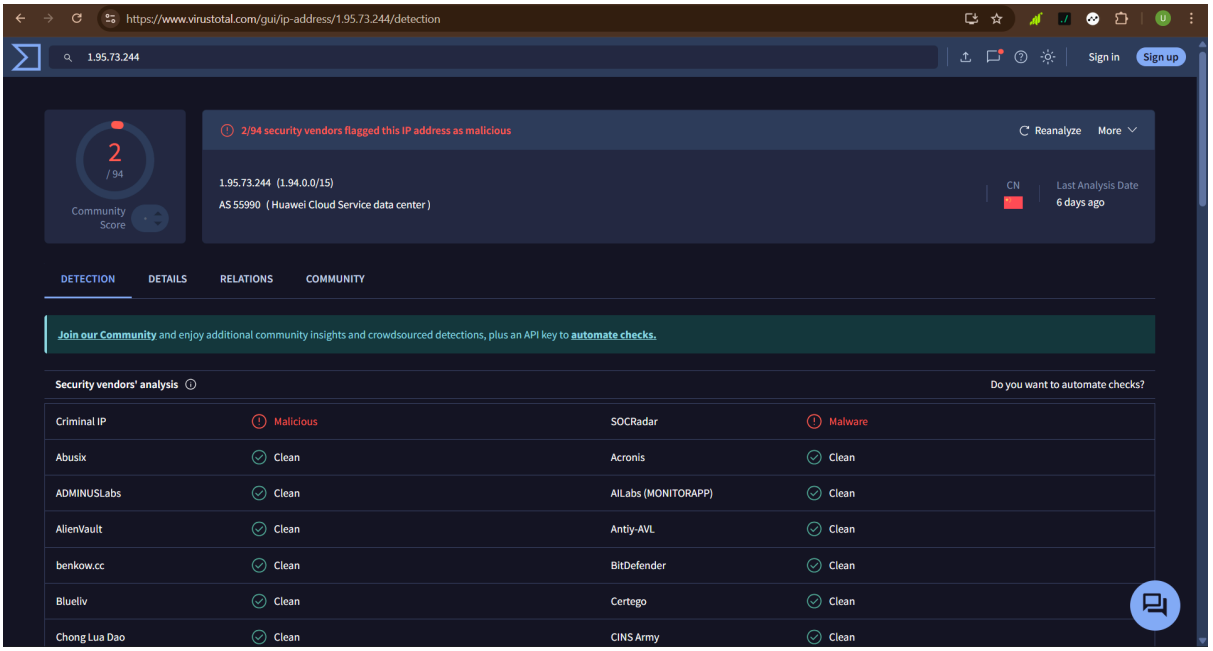
Attack Pattern Analysis:

- High-frequency bursts suggest automated brute force tooling
- Lack of successful login indicates password guessing rather than credential stuffing
- Consistent attack intervals indicate scripted execution

Threat Intelligence

- The IP was flagged as malicious on VirusTotal.

Figure 7: VirusTotal Result



Description: Shows IP reputation results.

Risk Assessment:

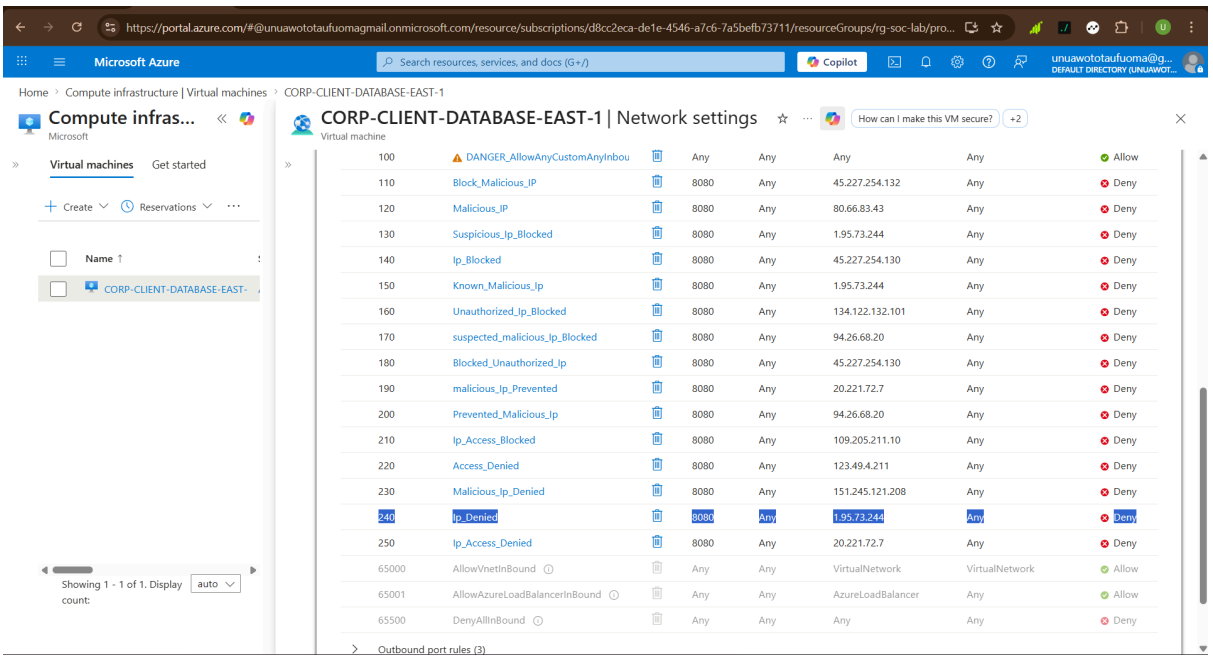
- High persistence
- High volume
- Targeted exposed service

13. INCIDENT RESPONSE

Actions taken:

- Verified malicious activity
- Blocked IP via NSG
- Monitored post-response activity

Figure 8: NSG Block Rule



Description: Shows NSG Rule Blocking Malicious IPs.

Outcome:

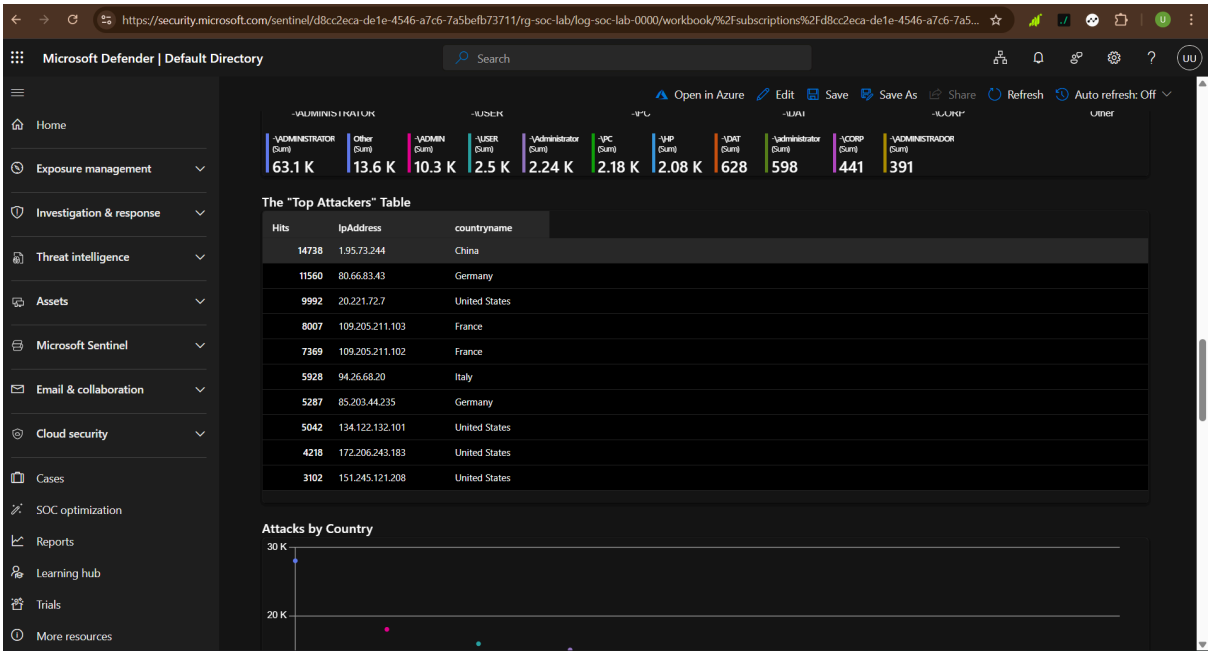
- Attack successfully contained
- No further activity observed

14. ADDITIONAL THREAT ACTORS

Attack sources identified from:

- China
- Germany
- United States
- France
- Italy

Figure 9: Top Attacker IP Table



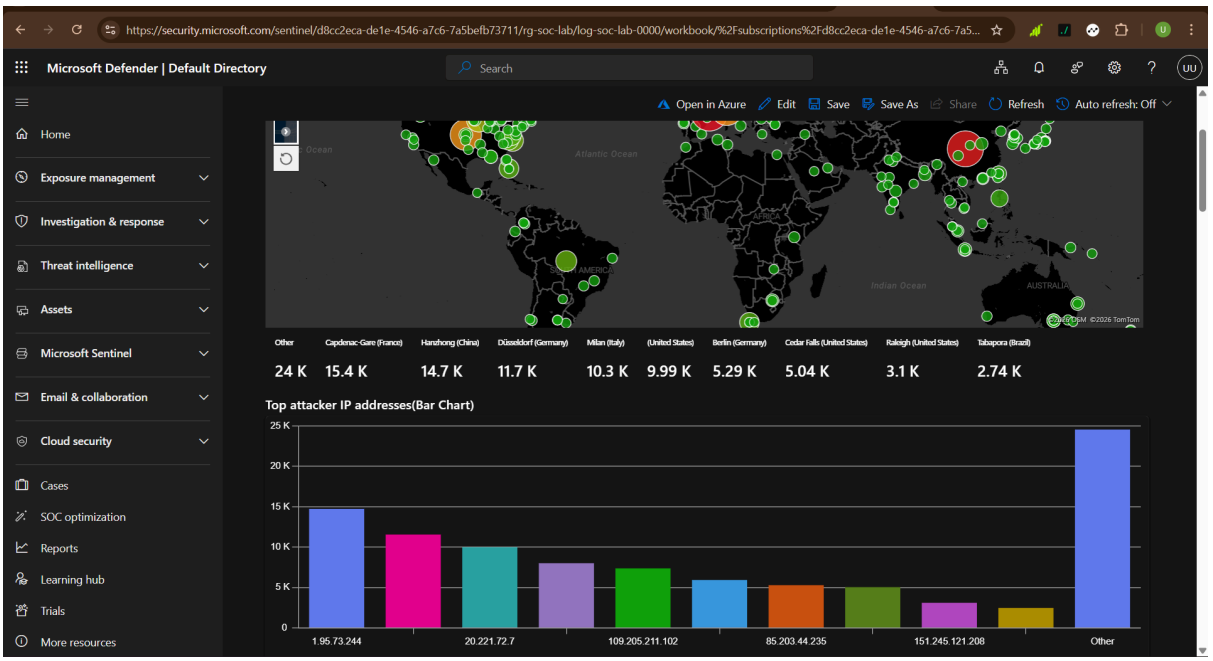
Description: Displays Attacks From Other Countries.

15. THREAT VISUALIZATION

Dashboard included:

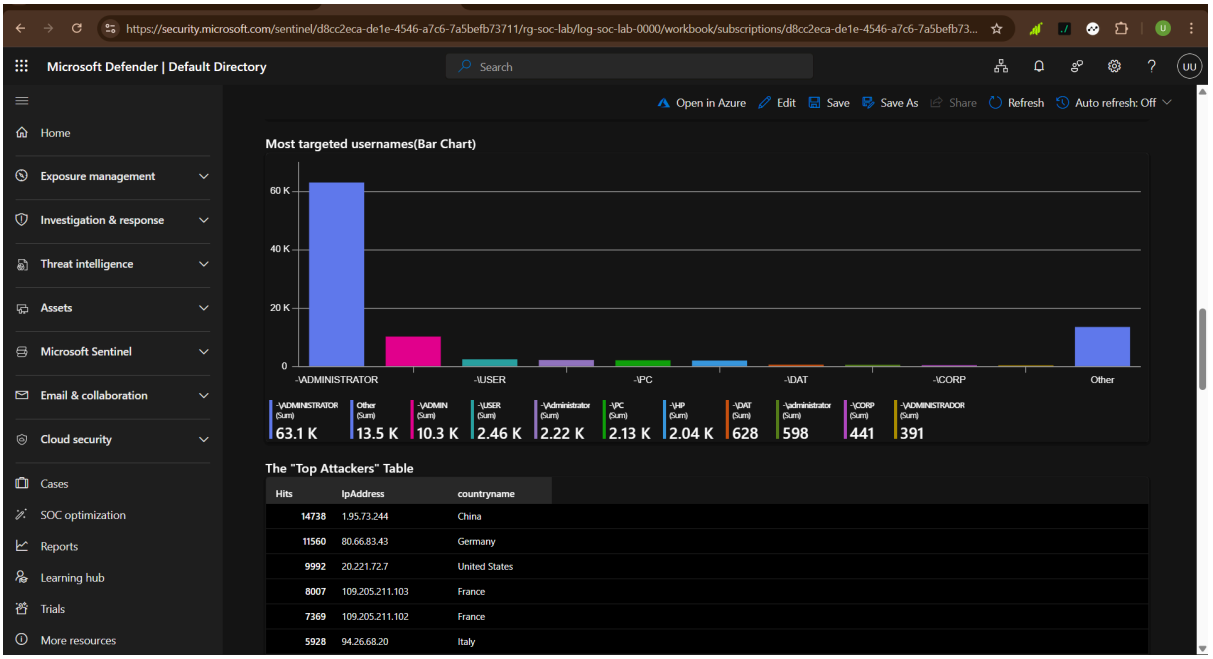
- Attack map
- Top attacker IPs
- Attack trends
- Live logs

Figure 10: Attack Map



Description: *Shows Geographic Distribution of Attacks.*

Figure 11: Workbook Dashboard



Description*: Shows Workbook With Most Targeted Usernames*

16. AUTOMATION

Automation was introduced to improve response efficiency.

Implemented Logic:

- *Trigger: High-severity incident detection*

- *Action: Alert grouping and prioritization*

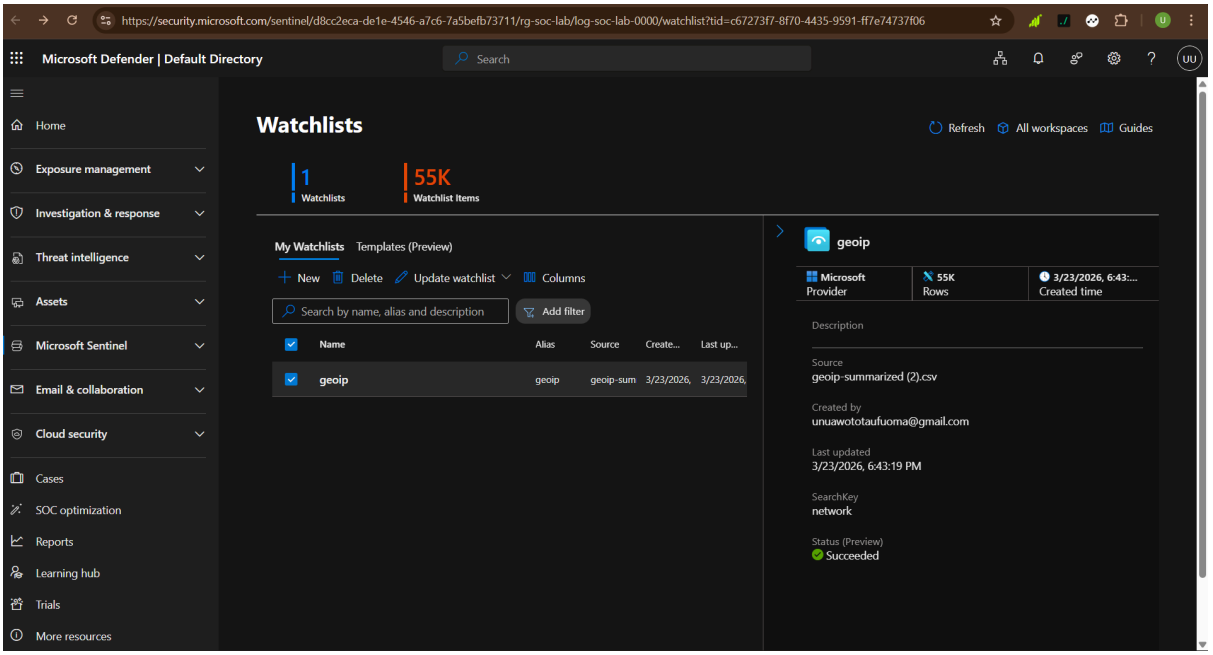
Future Improvements:

- *Automatic IP blocking using Logic Apps*
- *Integration with SOAR workflows*

17. THREAT INTELLIGENCE

- *Used VirusTotal for IP validation*
- *Created watchlists for malicious IP tracking*

Figure 12: Watchlist



Description: Shows Sentinel Watchlist For Geoip.

18. ALERT TUNING

- Adjusted detection thresholds
- Reduced false positives
- Improved alert accuracy

19. RISK & BUSINESS IMPACT ANALYSIS

If exploited in a real-world environment, exposed RDP services could lead to:

- Unauthorized system access
- Data exfiltration
- Deployment of ransomware
- Lateral movement within systems

In a business context, this could result in financial loss, reputational damage, and operational disruption.

20. RECOMMENDATIONS

- Disable public RDP access
- Implement Multi-Factor Authentication (MFA)
- Use Just-In-Time (JIT) access
- Apply IP allowlisting or geo-blocking
- Enforce strong password policies

21. KEY FINDINGS

- RDP services are heavily targeted immediately after exposure
- Attackers rely on automation tools
- Threats originate globally
- Continuous monitoring is essential

22. LESSONS LEARNED

- Detection logic must be tuned for accuracy
- Threat intelligence adds valuable context
- Proactive threat hunting enhances visibility
- Rapid response reduces risk exposure

23. CONCLUSION

This project successfully demonstrates practical SOC capabilities including detection engineering, incident investigation, threat hunting, and response execution.

The findings reinforce the importance of securing exposed services and highlight the effectiveness of SIEM-driven monitoring in detecting and mitigating threats.

This project reflects readiness for entry-level SOC analyst roles and provides a strong foundation for further advancement in cybersecurity.